
EECS 16A Designing Information Devices and Systems I

Homework 12

Submission Format

Your homework submission should consist of **one** file.

- `hw12.pdf`: A single PDF file that contains all of your answers (any handwritten answers should be scanned) as well as your IPython notebook saved as a PDF.

If you do not attach a PDF “printout” of your IPython notebook, you will not receive credit for problems that involve coding. Make sure that your results and your plots are visible. Assign the IPython printout to the correct problem(s) on Gradescope.

Submit the file to the appropriate assignment on Gradescope.

1. Reading Assignment

For this homework, please read Notes 21, 22, and 23 to learn about inner products, norms, trilateration, correlation, and least squares. It may also be helpful to recap Note 19 about op-amps, negative feedback, and common amplifiers designs. You are always encouraged to read beyond this as well.

- What does it mean for two vectors $\vec{x}$ and $\vec{y}$ to be orthogonal, in terms of their inner product?
- In trilateration, the distances between the beacons and the unknown location $\vec{x}$ involve quadratic terms of $\vec{x}$. What trick can we use to get a system of linear equations in $\vec{x}$?
- Suppose the signal $x[n]$ is only defined for timesteps $0, 1, \dots, 5$. For the purpose of computing linear cross-correlation, what value of $x[n]$ do we assume when n is a timestep out of the range: $0 \leq n \leq 5$ (e.g. $n = 6$ or $n = -1$)?

2. Mechanical Linear Correlation

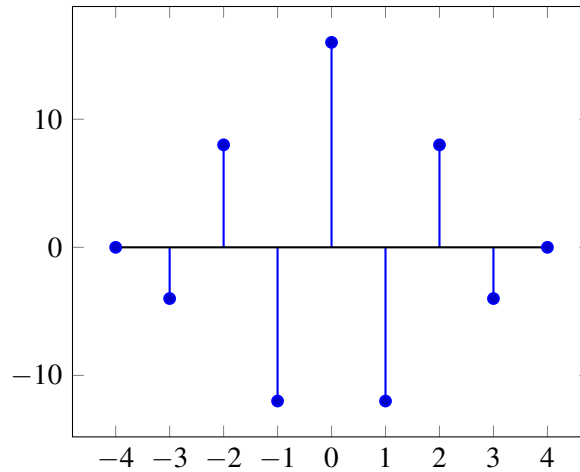
(Contributors: Michael Kellman, Avi Pandey, Linda Liu, Moses Won, Amanda Jackson, Gireeja Ranade, Lam Nguyen, Laura Brink, Sang Min Han, Spencer Kent, Urmita Sikder, Vijay Govindarajan, Raghav Anand)

Learning Goal: The objective of this problem is to understand how to compute the linear correlation between signals.

We recall that the linear correlation of signal $\vec{y}$ with signal $\vec{x}$ is given as:

$$\text{corr}_{\vec{x}}(\vec{y})[k] = \sum_{n=-\infty}^{\infty} x[n]y[n-k]$$

$\vec{s}_1[n]$	0	0	0	2	-2	2	-2	0	0	0								
$\vec{s}_1[n-3]$	0	0	0	0	0	0	2	-2	2	-2								
$\langle \vec{s}_1[n], \vec{s}_1[n-3] \rangle$	0	+	0	+	0	+	0	+	0	+	0	+	0	+	0	+	0	= -4



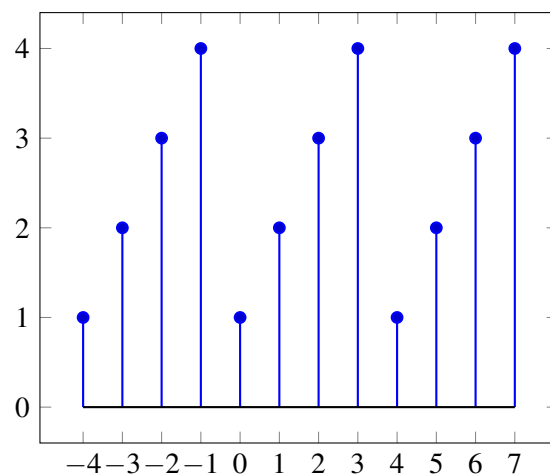
- (a) Using the procedure demonstrated above, compute $\text{corr}_{\vec{s}_1}(\vec{s}_2)[k]$, the linear cross-correlation of $\vec{s}_2$ with $\vec{s}_1$. Like the example, use tables like the one given below for $k = -3$ and plot the resulting correlation.

$\vec{s}_1[n]$	0	0	0	2	-2	2	-2	0	0	0
$\vec{s}_2[n+3]$	1	2	3	4	0	0	0	0	0	0
$\langle \vec{s}_1[n], \vec{s}_2[n+3] \rangle$										

- (b) Will the linear cross-correlation of $\vec{s}_2$ with $\vec{s}_1$ ($\text{corr}_{\vec{s}_1}(\vec{s}_2)[k]$) be the same as the cross-correlation of $\vec{s}_1$ with $\vec{s}_2$ ($\text{corr}_{\vec{s}_2}(\vec{s}_1)[k]$)? You can use the iPython notebook **prob12.ipynb** to figure this out. How are they related to each other?

Now, we will review the procedure to perform linear cross-correlation between one signal that is periodic with a period of 4 and another that is finite length and extended with zeros as in the previous parts. As an example, we will compute the linear correlation $\text{corr}_{\vec{p}_2}(\vec{s}_1)[k]$ between the periodic signal $\vec{p}_2$ (with period 4), formed by repeating $\vec{s}_2$, and the finite length signal $\vec{s}_1$ extended with zeros. The result will be a periodic signal with period 4.

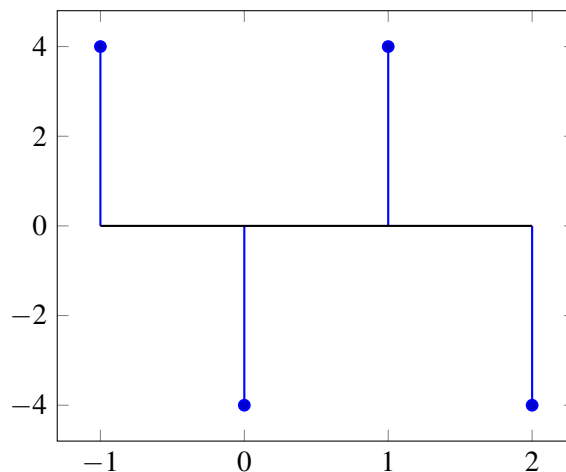
The periodic signal, $\vec{p}_2$, formed by repeating $\vec{s}_2$ is plotted below for indices -4 to 7. It is defined and non-zero for all indices from $-\infty$ to $+\infty$.



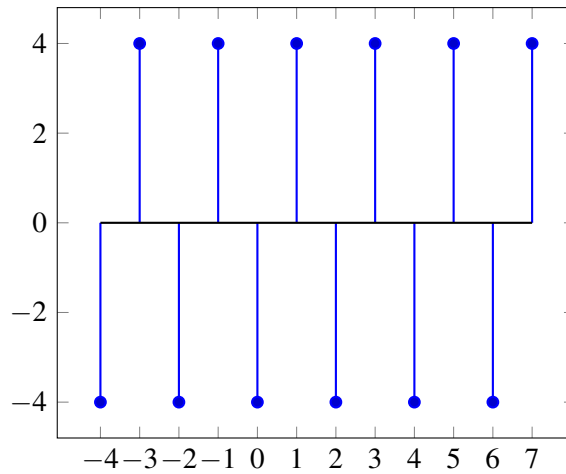
We compute one period of the result of the cross-correlation by starting at a shift of $k = -1$ and ending at a shift of $k = 2$.

$\vec{p}_2[n]$	2	3	4	1	2	3	4	1	2	3										
$\vec{s}_1[n+1]$	0	0	2	-2	2	-2	0	0	0	0										
$\langle \vec{p}_2[n], \vec{s}_1[n+1] \rangle$	0	+	0	+	8	+	-2	+	4	+	-6	+	0	+	0	+	0	+	0	=4
$\vec{p}_2[n]$	2	3	4	1	2	3	4	1	2	3										
$\vec{s}_1[n+0]$	0	0	0	2	-2	2	-2	0	0	0										
$\langle \vec{p}_2[n], \vec{s}_1[n+0] \rangle$	0	+	0	+	0	+	2	+	-4	+	6	+	-8	+	0	+	0	+	0	=-4
$\vec{p}_2[n]$	2	3	4	1	2	3	4	1	2	3										
$\vec{s}_1[n-1]$	0	0	0	0	2	-2	2	-2	0	0										
$\langle \vec{p}_2[n], \vec{s}_1[n-1] \rangle$	0	+	0	+	0	+	0	+	4	+	-6	+	8	+	-2	+	0	+	0	=4
$\vec{p}_2[n]$	2	3	4	1	2	3	4	1	2	3										
$\vec{s}_1[n-2]$	0	0	0	0	0	2	-2	2	-2	0										
$\langle \vec{p}_2[n], \vec{s}_1[n-2] \rangle$	0	+	0	+	0	+	0	+	0	+	6	+	-8	+	2	+	-4	+	0	=-4

The computed single period of the resulting linear cross correlation is plotted below.



The resulting linear cross correlation for shifts from $k = -4$ to $k = 7$ is plotted below.



- (c) Repeat the procedure described above to compute the correlation $\text{corr}_{\vec{p}_1}(\vec{s}_1)[k]$ between a periodic signal $\vec{p}_1$ (with period 4), formed by repeating $\vec{s}_1$, and the finite-length signal $\vec{s}_1$ extended with zeros. Like the example, evaluate tables like the one below for $k = -3$ for different shifts and plot a single period of the result.

$\vec{p}_1[n]$	-2	2	-2	2	-2	2	-2	2	-2	2
$\vec{s}_1[n+3]$	2	-2	2	-2	0	0	0	0	0	0
$\langle \vec{p}_1[n], \vec{s}_1[n+3] \rangle$										

3. Cauchy-Schwarz Inequality

(Contributors: Amanda Jackson, Avi Pandey, Gireeja Ranade, Michael Kellman, Panos Zarkos, Richard Liou, Vijay Govindarajan, Titan Yuan)

Learning Goal: The objective of this problem is to understand and prove the Cauchy-Schwarz inequality for real-valued vectors.

The Cauchy-Schwarz inequality states that for two vectors $\vec{v}, \vec{w} \in \mathbb{R}^n$:

$$|\langle \vec{v}, \vec{w} \rangle| = |\vec{v}^T \vec{w}| \leq \|\vec{v}\| \cdot \|\vec{w}\|$$

In this problem we will prove the Cauchy-Schwarz inequality for vectors in $\mathbb{R}^2$.

Take two vectors: $\vec{v} = r \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}$ and $\vec{w} = t \begin{bmatrix} \cos \phi \\ \sin \phi \end{bmatrix}$, where $r > 0, t > 0, \theta$, and ϕ are scalars. Make sure you understand why any vector in $\mathbb{R}^2$ can be expressed this way and why it is acceptable to restrict $r, t > 0$.

You may find it helpful to refer to trigonometric identities in your analysis.

- In terms of some or all of the variables r, t, θ , and ϕ , what are $\|\vec{v}\|$ and $\|\vec{w}\|$?
- In terms of some or all of the variables r, t, θ , and ϕ , what is $\langle \vec{v}, \vec{w} \rangle$?
- Show that the Cauchy-Schwarz inequality holds for any two vectors in $\mathbb{R}^2$. *Hint: consider your results from part (b). Also recall $-1 \leq \cos x \leq 1$ and use both inequalities.*
- Note that the inequality states that the inner product of two vectors must be less than or equal to the product of their magnitudes. What conditions must the vectors satisfy for the equality to hold? In other words, when is $\langle \vec{v}, \vec{w} \rangle = \|\vec{v}\| \cdot \|\vec{w}\|$?

4. GPS Receivers

(Contributors: Ava Tan, Aviral Pandey, Craig Schindler, Lam Nguyen, Michael Kellman, Moses Won, Terry Chern, Titan Yuan, Urmita Sikder, Vijay Govindarajan, Vasuki Narasimha Swamy)

Learning Goal: This problem will help to understand how GPS satellites transmit encoded signals to GPS receivers and how a receiver decodes the received signals and calculates the distance to the satellites using the signal propagation delays. It also shows how the GPS system is designed to be immune to noise.

The Global Positioning System (GPS) is a space-based satellite navigation system that provides location and time information in all weather conditions, anywhere on or near the Earth where there is an unobstructed line of sight to four or more GPS satellites. In this problem, we will understand how a receiver (e.g. your cellphone) can disambiguate signals from the different GPS satellites that are simultaneously received.

GPS satellites employ “spread-spectrum” technology (very similar to Code-division multiple access, CDMA, which is commonly used in cellphone transmissions) and a special coding scheme where each transmitter is assigned a code that serves as its “signature”.

Each GPS satellite uses a unique 1023 element long sequence as its “signature.” These codes used by the satellites are called “Gold codes,” and they have some special properties:

- The auto-correlation of a Gold code (cross-correlation with itself) is very **high** at the 0^{th} shift and very **low** at all other shifts.
- The cross-correlation between different Gold codes is very **low** at all shifts, i.e. different Gold codes are almost orthogonal to each other.

Gold codes are generated using a linear feedback shift register (LFSR). Understanding how this works is out of scope for the class, but you can read more about LFSR and CDMA if you are interested.

The important thing to know is that the Gold codes are 1023 element vectors where each element is either $+1$ or -1 , and that any Gold code is “almost orthogonal” to any other Gold code.

A receiver listening for signature transmissions from a satellite has copies of all of the different GPS satellites’ Gold codes. The receiver can determine how long it took for a particular GPS satellite’s signal to reach it by **taking the cross-correlation of the received signal with a satellite’s Gold code. (The Gold code is the signal which is shifted during cross-correlation — as discussed in lecture.)** The shift value (delay) that corresponds to distinct peaks (positive/negative) in the correlation determines the “propagation delay” between when the GPS satellite transmitted its signal and when the receiver received it. This time delay can then be converted into a distance (in the case of GPS, electromagnetic waves are used for transmissions, distance is equal to the speed of light multiplied by the time delay).

The GPS satellite is constantly transmitting its signature. In addition to identifying itself through its signature, it can also **“modulate” the signature** to communicate more information. Modulating a signature means multiplying the entire signature block by $+1$ or -1 , as shown in the figure.

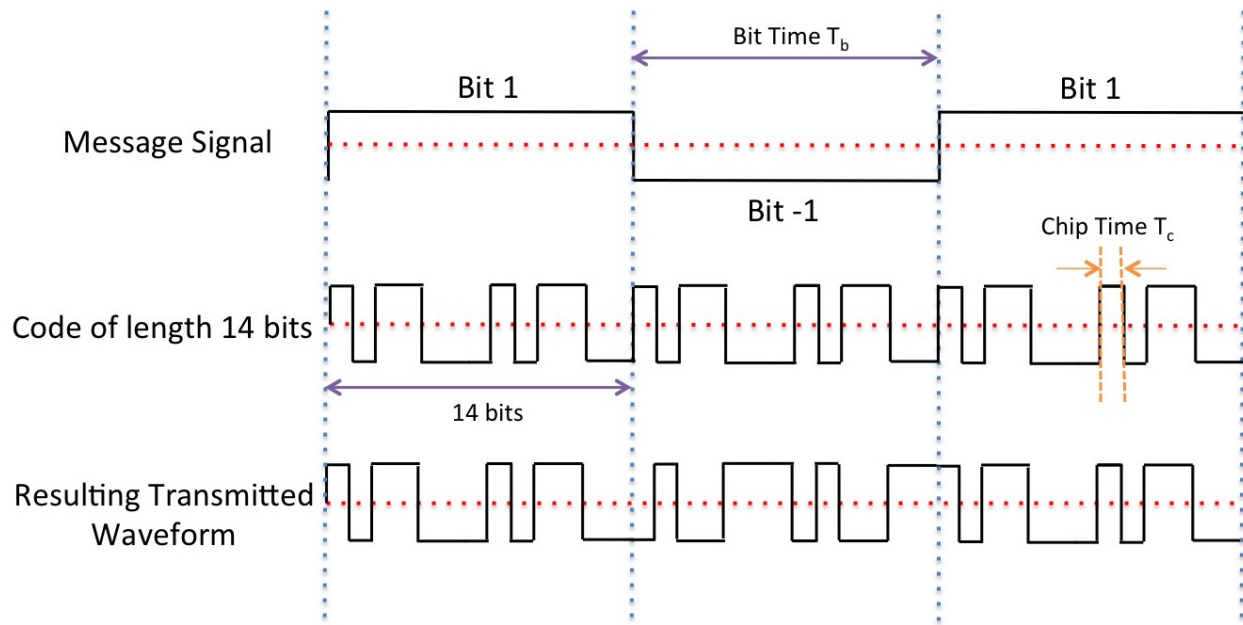
In the figure below, the signature is of length 14, i.e. it is made of 14 ± 1 symbols. T_c is the duration of one ± 1 symbol, and T_b is the duration of a whole signature. The figure shows 3 blocks of length 14 being transmitted. The message signal (made of $+1$ and -1 as well) multiplies the entire block of the signature, to give the resulting transformed waveform at the bottom of the figure. The message being transmitted in the figure is $[1 \ -1 \ 1]$. So to send these three symbols of message, we need to send 14×3 symbols of the gold code.

The waveform that is actually transmitted is the multiplication of the message signal with the signature signal. Now, when a receiver receives a signal, in addition to finding the time delay between transmission

and reception, the receiver will be able to decode the message by noting a very high correlation if the message bit is equal to 1, and a very negative correlation if the message bit is equal to -1 .

For the problem you will now do, $T_b = 1023T_c$. (In reality, transmissions can be much faster and have $T_b = 20 \times 1023 \times T_c$.)

You will use the ideas of linear correlation to figure out which of the satellites are transmitting.



For the purpose of this question we only consider 24 GPS satellites. Download the IPython notebook and the corresponding data files for the following questions:

Note: this code is calculation-heavy, and can take up to a few mins to run for each code block. Be patient!

The iPython code required for part (a) is already given for you. You only need to write codes for parts (b)-(g).

- Auto-correlate (i.e. cross-correlate with itself) the Gold code of satellite 10 and plot it. What do you observe? Use the helper function `array_correlation` in the notebook to perform correlation for all sub-parts of this problem. Try to understand what the helper function `array_correlation` is doing.
- Cross-correlate the Gold code of satellite 10 with satellite 13 and plot it. What do you observe?
- Consider a random signal, i.e. a signal that is not generated due to a specific code but is a random ± 1 sequence. A helper function `integernoise_generator` in the notebook will generate this for you. Cross-correlate it with the Gold code of satellite 10. What do you observe? What does this mean about our ability to identify satellites in the presence of random ± 1 noise?
- The signal actually received by a receiver will be the satellites' transmissions plus additive noise, and this need not be just noise that takes values ± 1 . Use the helper function `gaussiannoise_generator` in the notebook to generate a random noise sequence of length 1023, and compute the cross-correlation of this sequence with the Gold Code of satellite 10. What does this mean about our ability to identify satellites in the presence of real-valued noise?

For the next subparts of this problem, the received signals are corrupted by real-valued noise. Use the observation from this subpart for solving the rest of the question.

- (e) The receiver may receive signals from multiple satellites simultaneously, in which case the signals will all be added together. In addition, noise might be added to the signal. What are the satellites present in the received signal from `data1.npy`?

Use helper function `find_peak` in your code to help you figure out whether peak correlation values of magnitude greater than a pre-specified threshold value is present. Note that the threshold is 800 for this problem.

- (f) Let's assume that you can hear only one satellite, Satellite X, at the location you are in (though this never happens in reality). Let's also assume that this satellite is transmitting an unknown sequence of $+1$ and -1 of length 5 (after encoding it with the 1023 bit Gold code corresponding to Satellite X).

First, find out from `data2.npy` which satellite is transmitting using the same procedure that you used in part (e).

Next, find the 5 element sequence of ± 1 's that is being transmitted. To do this, you can observe the cross-correlation of the received signal from `data2.npy` with the Gold code of Satellite X and then visually find the peaks (positive /negative) and use these to understand the message.

- (g) **[OPTIONAL / NOT GRADED]**

Signals from different transmitters arrive at the receiver with different delays. We use these delays to figure out the distance between the satellite and receiver.

The signals from different satellites are superimposed on each other with different offsets at the start. What satellites are you able to see in `data3.npy`? Assume that all satellites begin transmission at time 0. What are the delays of all the satellites that are present? Assume you are told that all the satellites have the same message signal given by $[1 \ 1 \ -1 \ -1 \ -1]$.

5. Mechanical Trilateration

Trilateration is the problem of finding one's coordinates given distances from known location coordinates. For each of the following trilateration problems, you are given 3 positions and the corresponding distance from each position to your location. Find your location or possible locations. If a solution does not exist, state that it does not.

(a) $\vec{s}_1 = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$, $d_1 = 5$, $\vec{s}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, $d_2 = 2$, $\vec{s}_3 = \begin{bmatrix} -11 \\ 6 \end{bmatrix}$, $d_3 = 13$.

(b) $\vec{s}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $d_1 = 5\sqrt{2}$, $\vec{s}_2 = \begin{bmatrix} 10 \\ 0 \end{bmatrix}$, $d_2 = 5\sqrt{2}$, $\vec{s}_3 = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$, $d_3 = 5$.

(c) $\vec{s}_1 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, $d_1 = 5$, $\vec{s}_2 = \begin{bmatrix} 0 \\ -2 \end{bmatrix}$, $d_2 = 2$, $\vec{s}_3 = \begin{bmatrix} -12 \\ 5 \end{bmatrix}$, $d_3 = 12$.

6. Mechanical Projections

(Contributors: Michael Kellman)

Learning Goal: The objective of this problem is to practice calculating projection of a vector and the corresponding squared error.

(a) Find the projection of $\vec{b} = \begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix}$ onto $\vec{a} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$. What is the squared error between the projection and $\vec{b}$, i.e. $\|e\|^2 = \|\text{proj}_{\vec{a}}(\vec{b}) - \vec{b}\|^2$?

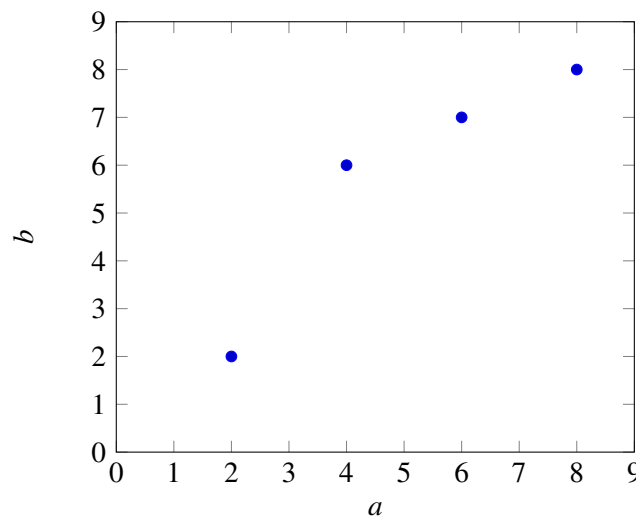
- (b) Find the projection of $\vec{b} = \begin{bmatrix} 1 \\ 4 \\ -5 \end{bmatrix}$ onto the subspace spanned by the vectors $\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$. What is the projection's squared error with $\vec{b}$, i.e. $\|e\|^2 = \|\text{proj}_{\vec{a}}(\vec{b}) - \vec{b}\|^2$?

7. Mechanical Least Squares

(Contributors: Adhyayan Narang, Aviral Pandey, Jiazhen Chen, Michael Kellman, Wahid Rahman)

Learning Goal:

The goal of this problem is to use least squares to fit different models (i.e. equations) to a data set. Depending on the model's number of parameters, the model will fit the data better or worse. A better model results in a lower squared error than a worse one. In part (a), we consider a linear model that contains a single slope parameter and intercepts the vertical axis at zero. In part (b), we consider a linear model with a possibly non-zero vertical axis intercept parameter, also known as an affine model.



- (a) Consider the above data points. Find the linear model of the form

$$\vec{a}x = \vec{b}$$

that best fits the data, where x is a scalar that minimizes the squared error

$$\|\vec{e}\|^2 = \left\| \begin{bmatrix} a_1 \\ \vdots \\ a_4 \end{bmatrix} x - \begin{bmatrix} b_1 \\ \vdots \\ b_4 \end{bmatrix} \right\|^2 = \|\vec{a}x - \vec{b}\|^2. \quad (1)$$

Note: By using this linear model, we are implicitly forcing the line to go through the origin.

You may use a calculator but show your work. Do not directly plug your numbers into IPython.

Note that we can model this linear model as a generic system $\mathbf{A}\vec{x} = \vec{b}$ where $\mathbf{A} = [\vec{a}]$ and $\vec{x} = x$. Once you've computed the optimal solution $\hat{x}$, compute the squared error between your model's prediction and the actual b values as shown in Equation 1. **Optional but recommended:** Plot the best fit line along with the data points to examine the quality of the fit. You may plot however you wish - one option is to use the helper code in the IPython notebook provided.

- (b) Now, let us consider an affine model for the same data, i.e. one with a non-zero vertical b -intercept. We believe we can get a better fit for the data by assuming an affine model of the form

$$ax_1 + x_2 = b,$$

for each point. Note that x_1 is the slope and x_2 is the b -intercept here. We can write this equation jointly for all the points using the vector notation below:

$$\vec{a}x_1 + \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} x_2 = \vec{b}.$$

Make sure you understand why a column vector of 1's is required above. In order to do this, we need to augment our $\mathbf{A}$ matrix from the previous part for the least squares calculation with a column of 1's (do you see why?), so that it has the form

$$\mathbf{A} = \begin{bmatrix} a_1 & 1 \\ \vdots & \vdots \\ a_4 & 1 \end{bmatrix}.$$

Set up a least squares problem to find the optimal x_1 and x_2 and compute the squared error between your model's prediction and the actual $\vec{b}$ values. Is it a better fit for the data? Provide a quantitative, numerical justification. **Optional:** Plot your affine model and examine qualitatively how close the best fit line is to the data points compared to part (a). You may plot however you wish - one option is to use the helper code in the IPython notebook provided.

- (c) **Prove the following theorem.**

Let $\mathbf{A} \in \mathbb{R}^{m,n}$. If $\vec{\hat{x}}$ is the solution to the least squares problem

$$\min_{\vec{x}} \left\| \mathbf{A}\vec{x} - \vec{b} \right\|^2,$$

then the error vector $\mathbf{A}\vec{\hat{x}} - \vec{b}$ is orthogonal to the columns of $\mathbf{A}$, i.e. show that: $\mathbf{A}^T(\mathbf{A}\vec{\hat{x}} - \vec{b}) = \vec{0}$.

Note: Trying to show individually that $\langle \vec{a}_i, \mathbf{A}\vec{\hat{x}} - \vec{b} \rangle = \vec{a}_i^T(\mathbf{A}\vec{\hat{x}} - \vec{b}) = 0$ for $i = 0, \dots, n$, where $\vec{a}_i$ is the i th column of $\mathbf{A}$ can be a bit tricky, however, stacking all of the $\vec{a}_i^T$ on top of each other and showing that $\mathbf{A}^T(\mathbf{A}\vec{\hat{x}} - \vec{b}) = \vec{0}$ is easier.

For this question, it is sufficient to prove that $\mathbf{A}^T(\mathbf{A}\vec{\hat{x}} - \vec{b}) = \vec{0}$.

Hint: Can you substitute $\vec{\hat{x}}$ using the least-squares formula?

8. Image Analysis

(Contributors: Amanda Jackson, Ava Tan, Aviral Pandey, Lydia Lee, Michael Kellman, Panagiotis Zarkos, Spencer Kent, Titan Yuan, Urmita Sikder)

Learning Goal: This problem introduces a method of fitting a non-linear model through a set of measured data points using the least squares method.

Applications in medical imaging often require an analysis of images based on the image's pixels. For instance, we might want to count the number of cells in a given biological sample. One way to do this is to take a picture of the cells and use the pixels to determine their locations and how many there are.

Automatic detection of shape is useful in image classification as well (e.g. consider a robot trying to find out autonomously where a mug is in its field of vision).

Let us focus back on the medical imaging scenario. You are interested in finding the exact position and shape of a cell in an image. You will do this by finding the **equation of the circle or ellipse** that bounds the cell relative to a given coordinate system in the image. Your collaborator uses edge detection techniques to find **a bunch of points that are approximately along the edge of the cell**. We assume that the origin of the coordinate system is in the center of the image with standard axes (x, y) and your collaborator gives you the following points that approximately bound the cell:

$(0.3, -0.69), (0.5, 0.87), (0.9, -0.86), (1, 0.88), (1.2, -0.82), (1.5, 0.64), (1.8, 0)$.

Recall that an equation of the form

$$a_1x^2 + b_1xy + c_1y^2 + d_1x + e_1y = 1$$

can be used to represent an ellipse (if $b_1^2 - 4a_1c_1 < 0$), and an equation of the form

$$a_1(x^2 + y^2) + d_1x + e_1y = 1$$

is a circle if $d_1^2 + e_1^2 + 4a_1 > 0$. Notice that the circle has fewer parameters.

- (a) How can you find the equation of a *circle* that surrounds the cell by fitting the data points? First, provide a setup and formulate a minimization problem to do this, i.e. a least squares problem minimizing the squared error $\|\mathbf{A}\vec{v} - \vec{b}\|^2$, where you attempt to find the **unknown coefficients** a_1 , d_1 , and e_1 from

your data points. Here your unknown vector $\vec{v} = \begin{bmatrix} a_1 \\ d_1 \\ e_1 \end{bmatrix}$. *Hint: The quantities $(x^2 + y^2)$, x , and y can be thought of as known values calculated from your data points.*

You do not need to simplify the numerical values for the matrix elements; just writing out the matrix with numerical expressions will suffice.

- (b) How can you find the equation of an ellipse (instead of a circle) that surrounds the cell? Provide a setup and formulate a minimization problem similar to that in part (a). Now the unknown vector $\vec{v}$ will be different from the earlier parts.

You do not need to simplify the numerical values for the matrix elements; just writing out the numerical expressions will suffice.

- (c) In the IPython notebook, write a short program that uses least-squares to fit a circle to the given points. A helper function `plot_circle` is provided. What is $\frac{\|\vec{e}\|}{N}$, where $\vec{e} = \mathbf{A}\vec{v} - \vec{b}$ and N is the number of data points? Plot your points and the best fit circle in IPython.

- (d) In the IPython notebook, write a short program that uses least-squares to fit an ellipse to the given points. A helper function `plot_ellipse` is provided. What is $\frac{\|\vec{e}\|}{N}$, where $\vec{e} = \mathbf{A}\vec{v} - \vec{b}$ and N is the number of data points? Now the unknown vector $\vec{v}$ will be different from the earlier parts. Plot your points and the best fit ellipse in IPython. How does this error compare to the one in the previous subpart? Which technique is better, and why?

9. Study Group Survey

We believe working collaboratively is one of the keys to success in this course! We want to learn more about your study group experience over the past semester, and ask you to fill out the following feedback form to help us better understand what collaboration looked like for you in EECS16A:

<https://forms.gle/pKXwv3wZfij64ui77>

To complete this part of the HW, you should fill out the survey and attach a screenshot indicating completion. Nothing else is required.

10. Homework Process and Study Group

Who did you work with on this homework? List names and student ID's. (In case you met people at homework party or in office hours, you can also just describe the group.) How did you work on this homework? If you worked in your study group, explain what role each student played for the meetings this week.